

Senior Thesis, Semester 1

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MA401

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Abstract

Continuing [CG16] and probably beginning [CG20]. The goal is to understand factorization algebras in the BV formalism.

Lecture 1: Translation invariant prefactorization algebras

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To start off, I will summarize [CG16] chapter 4.8, and we will move on to chapter 5 afterwards. I want to understand vertex algebras, so we will be reading the first 4 chapters of [FBZ04].

0.1 [CG16] 4.8 Translation-invariant prefactorization algebras

Following Costello and Gwilliam, we define the following. Given $x \in \mathbb{R}^n$ and $U \subseteq \mathbb{R}^n$, define $\tau_x U := U + x$.

Definition 0.1.1. Let $\mathcal{F}$ be a prefactorization on $\mathbb{R}^n$. Then $\mathcal{F}$ is *discretely translation invariant* if there are isomorphisms

$$\tau_x^f : \mathcal{F}(U) \cong \mathcal{F}(\tau_x U)$$

for all opens $U \subseteq \mathbb{R}^n$ and all points $x \in \mathbb{R}^n$. These isomorphisms satisfy the following coherence conditions.

1. We require that $\tau_x^f \circ \tau_y^f = \tau_{x+y}^f$ for all $x, y \in \mathbb{R}^n$.
2. For all disjoint opens $U_1, \dots, U_n \subseteq V \subseteq \mathbb{R}^n$, the diagram

$$\begin{array}{ccc} \mathcal{F}(U_1) \otimes \dots \otimes \mathcal{F}(U_n) & \xrightarrow{\tau_x^{\otimes n}} & \mathcal{F}(\tau_x U_1) \otimes \dots \otimes \mathcal{F}(\tau_x U_n) \\ \downarrow m & & \downarrow m \\ \mathcal{F}(V) & \xrightarrow{\tau_x} & \mathcal{F}(\tau_x V) \end{array}$$

commutes.

We generally write τ_x instead of τ_x^f , such that $\tau_x \mathcal{F}(U) \cong \mathcal{F}(\tau_x U)$.

We have seen a ton of these before. For example, observables of the free scalar field theory on $\mathbb{R}^n$ with or without mass. We could also do abelian Chern-Simons on $\mathbb{R}^3$.

This notion should be refined further. We have translation invariance for sections of $\mathcal{F}$, but it will be useful to have some translation invariance for the factorization product. In particular, m should vary smoothly based on the position of the open sets. We will introduce some notation to simplify.

Definition 0.1.2. A *degree k derivation* of a prefactorization algebra $\mathcal{F}$ on M valued in cochain complexes is a collection of maps $D_U : \mathcal{F}(U) \rightarrow \mathcal{F}(U)$ of cohomological degree k for each open $U \subseteq M$, with the property that for any finite collection $U_1, \dots, U_n \subseteq V$ of pairwise disjoint opens in M , and for any element $\alpha_i \in \mathcal{F}(U_i)$ for each i , the derivation acts by a Leibniz rule

$$D_V m_V^{U_1 \dots U_n} = \sum_i (-1)^{k(|\alpha_1| + \dots + |\alpha_{i-1}|)} m_V^{U_1 \dots U_n}(\alpha_1, \dots, D_{U_i} \alpha_i, \dots, \alpha_n).$$

Derivations of $\mathcal{F}$ form a cochain complex $\text{Der}^\bullet(\mathcal{F})$; in particular, they form a dg Lie algebra. Given a derivation D of degree k , the derivation dD of degree $k+1$ is defined as the bracket $(dD)_U = [d_U, D_U]$, for d_U the differential on $\mathcal{F}(U)$. Here we note that if D_U is degree k , then $D_U^n : \mathcal{F}(U)^n \rightarrow \mathcal{F}(U)[k]^n = \mathcal{F}(U)^{k+n}$. Thus composing with the differential yields a map to $\mathcal{F}(U)^{k+n+1}$, and thus dD is a derivation of degree $k+1$. The Lie bracket is defined by $[D, D']_U := [D_U, D'_U]$. Here recall we use the graded commutator

$$[f, g] = fg - (-1)^{|g|} gf.$$

Claim 0.1.3. $\text{Der}^\bullet(\mathcal{F})$ is indeed a dg Lie algebra.

Proof. We first show $d^2 = 0$. Let D be a derivation, and recall that since d_U is a differential on $\mathcal{F}$,

$$\begin{aligned}
(d^2 D)_U &= [d_U, [d_U, D_U]] \\
&= d_U[d_U, D_U] - (-1)^{|d_U, D_U|} [d_U, D_U] d_U \\
&= d_U[d_U, D_U] - (-1)^{|D|+1} [d_U, D_U] d_U \\
&= d_U[d_U, D_U] + (-1)^{|D|} [d_U, D_U] d_U \\
&= d_U^2 D_U - (-1)^{|D_U|} d_U D_U d_U + (-1)^{|D_U|} d_U D_U d_U - (-1)^{|D_U|} (-1)^{|D_U|} D_U d_U^2 \\
&= -(-1)^{|D_U|} d_U D_U d_U + (-1)^{|D_U|} d_U D_U d_U \\
&= 0
\end{aligned}$$

Now simply note that the graded commutator is a dg Lie bracket, so we only need to show that d is a derivation on the dg Lie bracket. Indeed, if D, D' are derivations, then by the graded Jacobi identity,

$$\begin{aligned}
(d[D, D'])_U &= [d_U, [D_U, D'_U]] \\
&= [[d_U, D_U], D'_U] + (-1)^{|d_U| |D_U|} [D_U, [d_U, D'_U]] = [(dD)_U, D'_U] + (-1)^{|D_U|} [D_U, (dD')_U].
\end{aligned}$$

Hence, d is a derivation for the dg Lie bracket. ■

Definition 0.1.4. An *action* of a dg Lie algebra $\mathfrak{g}$ on a prefactorization algebra $\mathcal{F}$ is a morphism $\mathfrak{g} \rightarrow \text{Der}^\bullet(\mathcal{F})$ of dg Lie algebras.

Given opens $U_1, \dots, U_n, V \subseteq \mathbb{R}^n$, let the set $\text{Disj}(U_1, \dots, U_k | V) \subseteq (\mathbb{R}^n)^k$ consist of all tuples $(x_1, \dots, x_k)$ such that each $\tau_{x_1} U_1, \dots, \tau_{x_k} U_k$ is still disjoint, and contained in V .

Let $\mathcal{F}$ be any discretely translation invariant factorization algebra on $\mathbb{R}^n$. Then the translation invariance gives us maps $\mathcal{F}(U_i) \rightarrow \mathcal{F}(\tau_{x_i} U_i)$, which assemble into a multilinear map

$$m_{x_1, \dots, x_k} : \mathcal{F}(U_1) \times \dots \times \mathcal{F}(U_k) \rightarrow \mathcal{F}(\tau_{x_1} U_1) \times \dots \times \mathcal{F}(\tau_{x_k} U_k) \rightarrow \mathcal{F}(V),$$

where the second map in the composite is the factorization product. Thus we can consider a map $\text{Disj}(U_1, \dots, U_k | V) \rightarrow \mathbb{H}\text{om}(\mathcal{F}(U_1), \dots, \mathcal{F}(U_k) | \mathcal{F}(V))$ given by $x \mapsto m_x$. Since $\mathcal{DVS}$ has an internal hom, so does $\text{Ch}_{\mathcal{DVS}}$, and thus it makes sense to talk about when this map is smooth.

Remark 0.1.5. We recall what the internal hom for differentiable vector spaces is. Let $V_1, \dots, V_n, W$ be differentiable vector spaces, and let M be a manifold. Let $\mathbf{C}^\infty(M, W)$ be the differentiable vector space $N \mapsto C^\infty(N \times M, W)$. A *smooth family* of multilinear maps $V_1 \times \dots \times V_n \rightarrow W$ parameterized by M is a map $V_1 \times \dots \times V_n \rightarrow \mathbf{C}^\infty(M, W)$. Intuitively, for every test manifold N , we have a map

$$V_1(N) \times \dots \times V_n(N) \rightarrow C^\infty(N \times M, W) = W(N \times M).$$

Let $m \in M$. Inclusion $i_m : N \cong N \times \{m\} \subseteq N \times M$ yields a pullback $i_m^* : W(N \times M) \rightarrow W(N)$. Postcomposing with the above map yields a map

$$V_1(N) \times \dots \times V_n(N) \rightarrow W(N \times M) \rightarrow W(N).$$

These maps are all natural in N , hence for all $m \in M$, we have a multilinear map $V_1 \times \dots \times V_n \rightarrow W$ of differentiable vector spaces. Now define $\mathbb{H}\text{om}(V_1, \dots, V_n | W)$ to be the differentiable vector space assigning a manifold M to all families of multilinear maps $V_1 \times \dots \times V_n \rightarrow W$. In the case $M = *$, each family is a singleton, hence $\mathbb{H}\text{om}(V_1, \dots, V_n | W)(*)$ is just multilinear maps $V_1 \times \dots \times V_n \rightarrow W$. This is exactly what we expect, since $E(*)$ recovers the underlying space of any differentiable vector space E .

Definition 0.1.6. A discretely translation invariant prefactorization algebra $\mathcal{F}$ on $\mathbb{R}^n$ is *smoothly translation invariant* if the following hold:

1. $m_{x_1, \dots, x_k}$ depends smoothly on $(x_1, \dots, x_k)$.
2. The prefactorization algebra $\mathcal{F}$ is equipped with an action of the abelian Lie algebra $\mathbb{R}^n$ of translations. If $v \in \mathbb{R}^n$, we denote the action maps by

$$\frac{d}{dv} : \mathcal{F}(U) \rightarrow \mathcal{F}(U).$$

3. The infinitesimal action is compatible with global translation invariance in the following sense. For $v \in \mathbb{R}^n$, denote by $v_i \in (\mathbb{R}^n)^k$ the vector obtained by inserting v into the i th slot. Then for all $\alpha_i \in \mathcal{F}(U_i)$,

$$\frac{d}{dv_i} m_{x_1, \dots, x_k}(\alpha_1, \dots, \alpha_k) = m_{x_1, \dots, x_k} \left(\alpha_1, \dots, \frac{d}{dv} \alpha_i, \dots, \alpha_k \right).$$

This definition makes good on the request that the factorization product depend smoothly on the position of the opens. I'm unsure why there is a smooth action by the Lie algebra $\mathbb{R}^n$.

For example, for the free scalar field theory on $\mathbb{R}$, the derivative ∂_x acts on the complex $\mathcal{E}$ of fields. It passes to the level of observables by asking that it be $\mathbb{R}[\hbar]$ -linear, and act as a derivation for the symmetric product. Since the symmetric product is precisely the factorization product, we see that this is an action of $\mathbb{R}$ on Obs^{cl} and Obs^q . We can generalize to theories on $\mathbb{R}^n$ by including all partials ∂_i .

When passing to the cohomology prefactorization algebra, we observe the same results. Note that the derivation determined by ∂_x on the cohomology pFA of observables on $\mathbb{R}$ corresponds to the Hamiltonian on the Weyl algebra.

From now on, whenever referring to a translation invariant prefactorization algebra, we always mean *smoothly* translation invariant.

There is a nice operadic reformulation of this definition.

Definition 0.1.7. Let Disks_n be the $\mathbb{R}_{>0}$ -colored operad whose n -multimorphisms are defined as follows. Given $r_1, \dots, r_n, s \in \mathbb{R}_{>0}$, the set $\text{Disks}_n(r_1, \dots, r_k | s) \subseteq (\mathbb{R}^n)^k$ is defined to be all k -tuples of points $x_i \in \mathbb{R}^n$ such that the balls $B_{r_1}(x_1), \dots, B_{r_k}(x_k)$ are all disjoint, and are all contained within $B_s(0)$. We refer to a multimorphism as a disk configuration. Composition laws are defined by translating the ball $B_s(0)$ to whatever point, and then inserting a disk configuration into this ball.

Remark 0.1.8. The operad Disks_n is like a colored version of the little n -disks operad, where we don't require the containing disk to have radius 1.

We write $\circ_i$ for the composition law in the i th slot, to simplify notation heavily. Thus

$$\circ_i : \text{Disks}_n(r_1, \dots, r_k | t_i) \times \text{Disks}_n(t_1, \dots, t_m | s) \rightarrow \text{Disks}_n(t_1, \dots, t_{i-1}, r_1, \dots, r_k, t_{i+1}, \dots, t_m | s).$$

Suppose that $\mathcal{F}$ is a translation invariant prefactorization algebra on $\mathbb{R}^n$. We can give $\mathcal{F}$ the structure of a Disks_n -algebra as follows. Write

$$\mathcal{F}_r := \mathcal{F}(B_r(0)),$$

and note that $\mathcal{F}_r$ is isomorphic to the sections of $\mathcal{F}$ over any ball of radius r by discrete translation invariance. Given a disk configuration $p \in \text{Disks}_n(r_1, \dots, r_k | s)$, write $m[p]$ for the factorization product

$$m[p] : \mathcal{F}_{r_1} \times \dots \times \mathcal{F}_{r_k} \rightarrow \mathcal{F}_s.$$

This is a multilinear map of differentiable vector spaces, and the map $p \mapsto m[p]$ is smooth since $\mathcal{F}$ is smoothly translation invariant. This yields a $\mathcal{D}\text{isks}_n$ -operad structure on $\mathcal{F}$ valued in differentiable cochains.

Let's consider some properties of $\text{Alg}_{\mathcal{D}\text{isks}_n}(\text{Ch}_{\mathcal{D}\text{VS}})$. Let N be a smooth manifold, and $r_i \in \mathbb{R}$. Then since $\mathcal{F}_{r_i}^{d_i}$ is a differentiable vector space, and hence a sheaf on Man , we have sections $\mathcal{F}_{r_i}^{d_i}(N) = C^\infty(N, \mathcal{F}_{r_i}^{d_i})$. Choose smooth maps $f_i : N \rightarrow \mathcal{F}_{r_i}^{d_i}$ for degrees d_i . Then the map

$$\begin{aligned} N \times \mathcal{D}\text{isks}_n(r_1, \dots, r_k | s) &\rightarrow \mathcal{F}_s \\ (x, p) &\mapsto m[p](f_1(x), \dots, f_k(x)) \end{aligned}$$

is smooth. Additionally, let $\sigma \in S_k$ be some permutation. Then there is an isomorphism

$$\sigma : \mathcal{D}\text{isks}_n(r_1, \dots, r_k | s) \cong \mathcal{D}\text{isks}_n(r_{\sigma(1)}, \dots, r_{\sigma(k)} | s)$$

defined in the evident way: $\sigma(x_1, \dots, x_k) = (x_{\sigma(1)}, \dots, x_{\sigma(k)})$.

Finally, note that operadic composition yields some coherence laws

$$m[q](\beta_1, \dots, \beta_{i-1}, m[p](\alpha_1, \dots, \alpha_k), \beta_{i+1}, \dots, \beta_\ell) = m[q \circ_i p](\beta_1, \dots, \beta_{i-1}, \alpha_1, \dots, \alpha_k, \beta_{i+1}, \dots, \beta_\ell).$$

We can also provide a co-operadic reformulation of this data. Define the $\mathbb{R}_{>0}$ colored co-operad by using the spaces $C^\infty(\mathcal{D}\text{isks}_n(r_1, \dots, r_k | s))$ and the completed tensor product $\widehat{\otimes}$. Since we know how to tensor a differentiable vector space with smooth maps on a manifold (via $C^\infty(N) \otimes V = C^\infty(N, V)$; we use these notations interchangeably), it makes sense to discuss an algebra over this co-operad in differentiable cochains.

The smoothness of the multiplication map $p \mapsto m[p]$ is rephrased as follows. Consider the map

$$\mathcal{F}_{r_1} \times \dots \times \mathcal{F}_{r_k} \rightarrow \mathcal{F}_s \otimes C^\infty(\mathcal{D}\text{isks}_n(r_1, \dots, r_k | s))$$

given by $(\alpha_1, \dots, \alpha_k) \mapsto m[-](\alpha_1, \dots, \alpha_k)$. Let

$$C^\infty(X, \mathbb{H}\text{om}(V_1, \dots, V_k | W))$$

denote the space of smooth multilinear maps $V_1 \times \dots \times V_k \rightarrow C^\infty(X, W)$. Note that there is a gluing map

$$\begin{aligned} C^\infty(X, \mathbb{H}\text{om}(V_1, \dots, V_k | W_i)) \times C^\infty(Y, \mathbb{H}\text{om}(W_1, \dots, W_\ell | T)) \\ \downarrow \circ_i \\ C^\infty(X \times Y, \mathbb{H}\text{om}(W_1, \dots, W_{i-1}, V_1, \dots, V_k, W_{i+1}, \dots, W_\ell | T)) \end{aligned}$$

defined as follows. If $f : V_1 \times \dots \times V_k \rightarrow C^\infty(X, W_i)$ and $g : W_1 \times \dots \times W_\ell \rightarrow C^\infty(Y, T)$, then we define

$$g \circ_i f : W_1 \times \dots \times W_{i-1} \times V_1 \times \dots \times V_k \times \dots \times W_\ell \rightarrow C^\infty(X \times Y, T)$$

by (here we take $w_a \in W_a$, and write $w_{<i}$ for $(w_1, \dots, w_{i-1})$, similarly for $w_{>i}$, and let $v \in V_1 \times \dots \times V_k$:

$$(g \circ_i f)(w_{<i}, v, w_{>i}) : (x, y) \mapsto g(w_{<i}, f(v)(x), w_{>i})(y).$$

We can now find elements

$$\mu(r_1, \dots, r_k | s) \in C^\infty(\mathcal{D}\text{isks}_n(r_1, \dots, r_k | s), \mathbb{H}\text{om}_{\mathcal{D}\text{VS}}(\mathcal{F}_{r_1}, \dots, \mathcal{F}_{r_k} | \mathcal{F}_s))$$

defined by

$$p \mapsto m[p]$$

for $m[p] : \mathcal{F}_{r_1} \times \dots \times \mathcal{F}_{r_n} \rightarrow \mathcal{F}_s$, with the following properties.

1. There is a natural differential arising from the differentials on $\mathcal{F}_{r_i}$, and these elements are closed under it. The differential is defined using the internal on the internal hom of cochains. If $V_1, \dots, V_k, W$ are differentiable cochains, then $\underline{\text{Hom}}(V_1, \dots, V_k | W)$ is defined by taking degree q elements to be multilinear maps of degree q . The differential is then defined in the usual way:

$$(\text{d}\phi)(v_1, \dots, v_k) = \text{d}_W(\phi(v_1, \dots, v_k)) - (-1)^{|\phi|} \sum_{i=1}^k (-1)^{|v_1| + \dots + |v_{i-1}|} \phi(v_1, \dots, \text{d}v_i, \dots, v_k).$$

Recall that the differential on tensors acts as a derivation. Thus this is just

$$\text{d}(\phi) = \text{d}_W \circ \phi - (-1)^{|\phi|} \phi \circ \text{d}_{V_1 \otimes \dots \otimes V_k}.$$

2. If $\sigma \in S_k$ is a permutation, then

$$\sigma_* \mu(r_1, \dots, r_k | s) = \mu(r_{\sigma(1)}, \dots, r_{\sigma(k)} | s),$$

where

$$\begin{array}{c} C^\infty(\text{Disks}_n(r_1, \dots, r_k | s), \mathbb{H}\text{om}_{\mathcal{DVS}}(\mathcal{F}_{r_1}, \dots, \mathcal{F}_{r_k} | \mathcal{F}_s)) \\ \downarrow \sigma_* \\ C^\infty(\text{Disks}_n(r_{\sigma(1)}, \dots, r_{\sigma(k)} | s), \mathbb{H}\text{om}_{\mathcal{DVS}}(\mathcal{F}_{r_{\sigma(1)}}, \dots, \mathcal{F}_{r_{\sigma(k)}} | \mathcal{F}_s)) \end{array}$$

is the natural isomorphism.

3. Let $\circ_i$ be the composition map on Disks_n . Then there is an induced map $\circ_i^*$, and

$$\sigma_i^* \mu(t_1, \dots, t_{i-1}, r_1, \dots, r_k, t_{i+1}, \dots, t_\ell) = \mu(r_1, \dots, r_k | t_i) \circ_i \mu(t_1, \dots, t_\ell | s).$$

These elements equip $\mathcal{F}$ with the structure of a Disks_n -algebra, as stated before.

For example, consider the free scalar field on $\mathbb{R}^2$ with no mass. Recall that given a Green's function G , we have an isomorphism $W_U : \text{Obs}^{cl}(U)[\hbar] \rightarrow \text{Obs}^q(U)[\hbar]$ for all opens U , given by $e^{\hbar \partial_G}(\alpha)$. This yields an isomorphism of prefactorization algebras, when $\text{Obs}^{cl}[\hbar]$ is equipped with the deformed factorization product

$$\alpha \star_{\hbar} \beta = W_{U \sqcup V}^{-1}(W_U \alpha \cdot W_V \beta).$$

Here, $\cdot$ is the factorization product on Obs^q , and $U, V \subseteq M$ are disjoint opens. This allows us to explicitly describe the product maps

$$\mu(r_1, \dots, r_k | s) : \mathcal{F}_{r_1} \otimes \dots \otimes \mathcal{F}_{r_k} \rightarrow C^\infty(\text{Disks}_n(r_1, \dots, r_k | s), \mathcal{F}_s).$$

Here, $\mathcal{F} = \text{Obs}^q$, or $(\text{Obs}^{cl}[\hbar], \star_{\hbar})$. Given radii r_1, r_2 and compactly supported smooth functions α_1, α_2 on $B_{r_1}(0)$ and $B_{r_2}(0)$, we define $\mu(\alpha_1, \alpha_2)$ to be the function assigning a disk configuration (x_1, x_2) to the element

$$\mu_{r_1, r_2}^s(\alpha_1, \alpha_2)(x_1, x_2) = \tau_{x_1} \alpha_1 \star_{\hbar} \tau_{x_2} \alpha_2.$$

Using the Green's function $G(x, y) = -\frac{1}{2\pi} \ln |x - y|$, which works on $\mathbb{R}^2$ when $m = 0$, then this equals

$$(\tau_{x_1} \alpha_1) \cdot (\tau_{x_2} \alpha_2) - \hbar \int_{u_1, u_2 \in \mathbb{R}} \alpha_1(u_1 + x_1) \alpha_2(u_2 + x_2) \ln |u_1 - u_2|.$$

To see this, recall that ∂_G contracts with G , and is trivial on linear observables. Thus $\partial_G(\alpha_i) = 0$, hence $e^{\hbar\partial_G}\alpha_i = \alpha_i$. Thus we only have $W^{-1}(\tau_{x_1}\alpha_1 \cdot \tau_{x_2}\alpha_2) = e^{\hbar\partial_G}(\alpha_1\alpha_2)$. We *can* contract with a quadratic observable, and contraction simply gives

$$\partial_G(\alpha\beta) = \int \alpha(u_1)G(u_1, u_2)\beta(u_2).$$

Thus we see the 0th and 2nd order terms of the exponential, and higher order terms vanish. I don't know where the $-\frac{1}{2\pi}$ has gone.

Lecture 2: Vertex algebras

Wed 23 Sep 2026 23:19

0.2 Definition of vertex algebras, [FBZ04]

Definition 0.2.1. Let R be a $\mathbb{C}$ -algebra. An R -valued formal power series (or formal distribution) in variables $z_1, \dots, z_n$ is an arbitrary (finite or infinite) series

$$A(z_1, \dots, z_n) = \sum_{i_1 \in \mathbb{Z}} \cdots \sum_{i_n \in \mathbb{Z}} A_{i_1, \dots, i_n} z_1^{i_1} \cdots z_n^{i_n}$$

where the coefficients $A_{i_1, \dots, i_n} \in R$. These series form a vector space, denoted $R[[z_1^{\pm 1}, \dots, z_n^{\pm 1}]]$.

The vector space $R[[z_1^{\pm 1}, \dots, z_n^{\pm 1}]]$ is not a $\mathbb{C}$ -algebra, since the product of two formal power series does not always make sense: the product must still have coefficients A , but when you take the product PQ of two formal power series, these coefficients themselves become infinite series formed out of the coefficients of P and Q , which need not converge.

However, this is since these formal power series act in the same variables, so we end up collecting infinite terms. If we have $P \in R[[z_1^{\pm 1}, \dots, z_n^{\pm 1}]]$ and $Q \in R[[w_1^{\pm 1}, \dots, w_n^{\pm 1}]]$, then we can get a well-defined product PQ in $R[[z_1^{\pm 1}, \dots, z_n^{\pm 1}, w_1^{\pm 1}, \dots, w_n^{\pm 1}]]$. Additionally, the product with a Laurent polynomial (i.e. all but finitely many coefficients vanish) is clearly always well-defined.

Definition 0.2.2. Given a formal power series in one variable f , we define the residue $\text{Res } f(z)dz = a_{-1}$.

Note that if $R = \mathbb{C}$, then this agrees with the usual notion of residue for a function f :

$$\text{Res } f(z)dz = \frac{1}{2\pi i} \oint f(z)dz.$$

Given such a formal power series $f \in \mathbb{C}[[z^{\pm 1}]]$, we obtain a linear functional on Laurent polynomials $\mathbb{C}[z, z^{-1}]$ via

$$g \mapsto \text{Res } f(z)g(z)dz.$$

This generalizes to formal power series in multiple dimensions, and rational functions in those dimensions. For this reason, we often call these *formal distributions*.

For example, define the *formal dirac delta* in two variables z, w as

$$\delta(z - w) = \sum_{m \in \mathbb{Z}} z^m w^{-m-1}.$$

Writing this as a series in 2 variables, the coefficients are $a_{nm} = \delta_{m,-n-1}$:

$$\delta(z - w) = \sum_m \sum_n \delta_{m,-n-1} z^n w^m.$$

Hence, the coefficients are supported on the diagonal $m + n = -1$, so given a formal power series A in 1 variable w , we can multiply

$$A(w)\delta(z - w) = \sum_{m,n \in \mathbb{Z}} A_{m+n+1} z^m w^n.$$

Thus we see

$$A(w)\delta(z - w) = A(z)\delta(z - w).$$

We thus obtain from this fact that

$$(z - w)\delta(z - w) = 0,$$

and induction yields

$$(z - w)^{n+1} \partial_w^n \delta(z - w) = 0.$$

I want to prove this fact since it seems nontrivial. Assume the inductive hypothesis. Then

$$(z - w)^{n+2} \partial_w^{n+1} \delta(z - w) = (z - w)^{n+2} \partial_w \partial_w^n \delta(z - w).$$

We would like to pull out this ∂_w to use the inductive hypothesis. Since derivatives act by a Leibniz rule, we have

$$\partial_w ((z - w)^{n+2} \partial_w^n \delta(z - w)) = -(n+2)(z - w)^{n+1} \partial_w^n \delta(z - w) + (z - w)^{n+2} \partial_w^{n+1} \delta(z - w).$$

Reorganizing yields

$$\partial_w ((z - w)^{n+2} \partial_w^n \delta(z - w)) + (n+2)(z - w)^{n+1} \partial_w^n \delta(z - w) = (z - w)^{n+2} \partial_w^{n+1} \delta(z - w).$$

However,

$$(z - w)^{n+1} \partial_w^n \delta(z - w) = 0$$

by the inductive hypothesis. Hence

$$\partial_w ((z - w)^{n+2} \partial_w^n \delta(z - w)) = (z - w)^{n+2} \partial_w^{n+1} \delta(z - w).$$

We are now essentially done. Write

$$(z - w)^{n+2} \partial_w^n \delta(z - w) = (z - w)(z - w)^{n+1} \partial_w^n \delta(z - w) = (z - w)0 = 0$$

by the inductive hypothesis, and thus

$$\partial_w(0) = (z - w)^{n+2} \partial_w^{n+1} \delta(z - w).$$

By the method of induction, we are done.

These properties yield an abstract characterization of δ and its derivatives, due to Kac.

Lemma 0.2.3. *Let $f(z, w)$ be a formal power series in $R[[z^{\pm 1}, w^{\pm q}]]$ satisfying $(z - w)^N f(z, w) = 0$ for some $N \geq 0$. Then*

$$f(z, w) = \sum_{i=0}^{N-1} g_i(w) \partial_w^i \delta(z - w), \quad g_i(w) \in R[[w^{\pm 1}]].$$

Moreover, the g_i 's are unique.

Proof. Since $(z - w)\delta(z - w) = 0$, we see that for any $N \geq 1$, we obtain

$$(z - w)^N \delta(z - w) = (z - w)^{N-1} (z - w) \delta(z - w) = 0.$$

We must show

$$(z - w)^N \sum_{i=0}^{N-1} g_i(w) \partial_w^i \delta(z - w) = 0.$$

This is immediate: simply distribute in the $(z - w)^N$, commute it with $g_i(w)$ (both of these can be done since $(z - w)^N$ is Laurent), and we see that every term vanishes. Noting here that $0 \leq i \leq N - 1$, we have

$$\begin{aligned} (z - w)^N g_i(w) \partial_w^i \delta(z - w) &= g_i(w) (z - w)^N \partial_w^i \delta(z - w) \\ g_i(w) (z - w)^{N-i-1} (z - w)^{i+1} \partial_w^i \delta(z - w) &= 0. \end{aligned}$$

Thus, we have proven all series of this form satisfy the given property.

Now we prove all series satisfying the given property are of this form. Write

$$f(z, w) = \sum_{n, m \in \mathbb{Z}} f_{n, m} z^n w^m.$$

Define $(\Delta f)_{n, m} := f_{n-1, m} - f_{n, m-1}$. Then we claim $(\Delta^N f)_{n, m} = 0$ for all n, m . Here we interpret Δ as an operator on formal distributions which produces the formal power series whose coefficients are as defined. We prove $(\Delta^N f)_{n, m} = 0$. To do this, we proceed inductively. First, note that

$$\begin{aligned} (z - w)f(z - w) &= (z - w) \sum_{n, m \in \mathbb{Z}} f_{n, m} z^n w^m = \sum_{n, m \in \mathbb{Z}} f_{n, m} z^{n+1} w^m - f_{n, m} z^n w^{m+1} \\ &= \sum_{n, m \in \mathbb{Z}} f_{(n+1)-n, m} z^{n+1} w^m - f_{n, (m+1)-1} z^n w^{m+1} = \sum_{n, m \in \mathbb{Z}} (\Delta f)_{n, m} z^n w^m = \Delta f. \end{aligned}$$

Now assume the relevant inductive hypothesis for some n . We compute

$$(z - w)^{n+1} f(z - w) = (z - w) (\Delta^n f)(z, w) = \Delta (\Delta^n f)(z, w) \Delta^{n+1} f.$$

By induction, we in particular obtain $\Delta^N f = (z - w)^N f(z - w)$. Since the latter vanishes by hypothesis, we find that all coefficients of $\Delta^N f$ vanish.

Note that the equations $(\Delta^N f)_{n, m}$ can be reduced to equations on a particular diagonal. In particular, we can fix k , and consider coefficients on the diagonal $f_{k, p-k}$. These must all sum to 0. Hence we can define $f^{(p)}$ to be supported on this diagonal, meaning $f_{n, m}^{(p)} \neq 0$ if and only if $n + m = p$. Thus

$$f(z, w) = \sum_{p \in \mathbb{Z}} f^{(p)}(z, w).$$

On the p th diagonal, the equation $(\Delta^N f)_{n,m} = 0$ becomes a difference equation of N functions using the binomial theorem in a sort of way:

$$(\Delta^N f)_{n,m} = \sum_{j=0}^N (-1)^j \binom{N}{j} f_{n-N+j, m-j}.$$

■

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